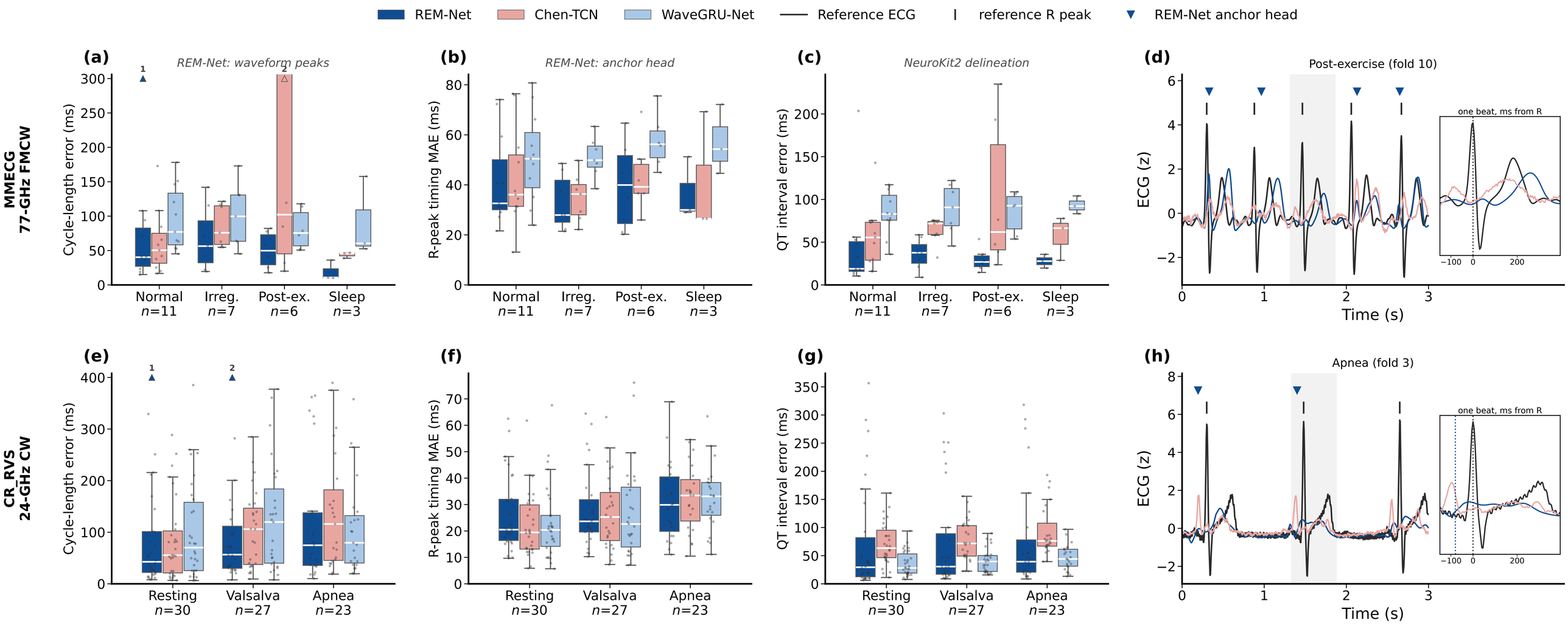

REM-Net
Chen-TCN
WaveGRU-Net
Reference ECG
reference R peak
REM-Net anchor head
MMECG
77-GHz FMCW
CR_RVS
24-GHz CW
(a)
REM-Net: waveform peaks
Cycle-length error (ms)
(b)
REM-Net: anchor head
R-peak timing MAE (ms)
(c)
NeuroKit2 delineation
QT interval error (ms)
(d)
Post-exercise (fold 10)
ECG (z)
Time (s)
one beat, ms from R
(e)
(f)
(g)
(h)
Apnea (fold 3)
Normal n=11
Irreg. n=7
Post-ex. n=6
Sleep n=3
Resting n=30
Valsalva n=27
Apnea n=23